EE2004 Digital Signal Processing

Tutorial 3 Answer Key

1. (a) $X(z) = \dfrac{-1.2\,z^{-1}}{1 - 2.8\,z^{-1} + 1.6\,z^{-2}}$, RoC: $0.8 < |z| < 2$,
   (b) $X(z) = \dfrac{2 - 5\,z^{-1}}{1 - 5\,z^{-1} + 6\,z^{-2}}$, RoC: $2 < |z| < 3$,
   (c) $X(z) = \dfrac{-0.1\,z^{-1}}{1 - 1.7\,z^{-1} + 0.72\,z^{-2}}$, RoC: $0.8 < |z| < 0.9$,
   (d) $X(z) = \dfrac{-4 + 2.8\,z^{-1}}{1 - 1.2\,z^{-1} + 0.32\,z^{-2}}$, RoC: $|z| < 0.4$,
   (e) $X(z) = \dfrac{4 - 2.8\,z^{-1}}{1 - 1.2\,z^{-1} + 0.32\,z^{-2}}$, RoC: $|z| > 0.8$,
   (f) $X(z) = \dfrac{0.75}{1.25 - 0.5\,(z + z^{-1})}$, RoC: $0.5 < |z| < 2$.

2. $X(z) = \dfrac{1}{1 - e^{j\omega_0} z^{-1}}$, RoC: $|z| > 1$.

3. (a) $X(z) = \dfrac{1 - \cos\omega_0\, z^{-1}}{1 - 2\,\cos\omega_0\, z^{-1} + z^{-2}}$, RoC: $|z| > 1$, (b) $X(z) = \dfrac{\sin\omega_0\, z^{-1}}{1 - 2\,\cos\omega_0\, z^{-1} + z^{-2}}$, RoC: $|z| > 1$.

4. (a) $|z| < e^{-2}$, left-sided (a) $e^{-2} < |z| < 2$, two-sided (c) $|z| > 2$, right-sided

5. (a) 1; RoC: finite z-plane, (b) z^{-n_0}; RoC: finite z-plane except $z = 0$, (c) z^{n_0}; RoC: finite z-plane except $z = \infty$.

6. $X(z) = \dfrac{1}{1 - z^{-1}}$, RoC: $|z| > 1$.

7. $H(z) = (1 - z^{-20})(1 - 0.8\,z^{-1})$, RoC: finite z-plane except $z = 0$.

8. $X(z) = \dfrac{1}{1 - 2r\,\cos\theta\, z^{-1} + r^2\, z^{-2}}$, RoC: $|z| > r$.